NIAGARA SUCCESSION SIGNAL

Week 2 Documentation

Data Source Research, Evaluation & Planning

DataraFlow Founder Challenge 2026

Prepared by: **Joyce — Team Lead**

Project focus: Small-business succession risk across Niagara Region, Ontario

Week 2 Status: **COMPLETE**

1. Week 2 Purpose

This week, our goal was to understand what data is available for the Niagara Succession Signal project before we begin collecting, cleaning, combining or modelling data. I wanted the team to move beyond simply finding links. We needed to understand what each source contains, how it could help our research question, what it cannot tell us, and whether it should move forward into the next stage.

> **Main Week 2 question:** What data is available, what can it help us answer, and what are its limitations?

2. Team Research Contributions

Team members	Research area	Main contribution / decision
Stephen & Mano	Niagara Employment Inventory (NEI) & Niagara Open Data	Identified the Consolidated NEI as a key local business source. Reported about 98,065 records and fields relating to municipality, NAICS, sector/industry, business opening year and employee-size range. Direct succession variables were not observed in the public attributes inspected. Keep as Priority A.
Oluwatosin & Ohonba	Retirement, succession planning, exit timelines and barriers	Found Canadian/Ontario succession-related material, including Statistics Canada and CFIB resources. The major limitation was geography: much of the succession evidence was not Niagara-specific. Retain relevant datasets/research selectively.
Damilare & Lucky	Statistics Canada Business Dynamics / Entries & Exits	Identified business entrant, incumbent, exit, survival and firm-size information. Correctly noted that a business exit does not reveal whether the cause was failure, sale, retirement or lack of succession. Keep selected tables for benchmarks and context.
Precious & Ruth	Statistics Canada Census & Demographics	Identified age, class-of-worker, self-employment, industry/occupation and geographic information useful for Niagara context. Self-employment is a proxy and must not automatically be treated as business ownership. Keep selected tables as Priority A.
Joyce — Team Lead	Coordination, review and documentation	Coordinated source research, reviewed updates, separated direct/proxy/context variables, consolidated data gaps, organised acquisition priorities, documented risks and licences, prepared Week 3 acquisition/audit structures, and aligned the work with challenge deliverables.

3. Main Data Sources Selected

Source	Week 2 decision	Planned role
Consolidated Niagara Employment Inventory (NEI)	Priority A — Keep	Main local business structure: municipality, industry/NAICS, employee size and other confirmed business characteristics.
Statistics Canada — Canadian Business Counts (33-10-1176-01)	Priority A — Keep	Business-count benchmarks by available geography, industry and employment size.
Statistics Canada — Business Dynamics by Industry (33-10-0087-01)	Priority A — Keep	Historical entry/exit patterns by industry; exit is not treated as succession failure.
Statistics Canada — Business Dynamics by Firm Size (33-10-0088-01)	Priority A — Keep	Historical business dynamics by firm size; mainly benchmark/context.
Statistics Canada — Census Age (98-10-0022-01)	Priority A — Keep	Niagara demographic age structure; population age is not business-owner age.

Source	Week 2 decision	Planned role
Statistics Canada — Class of Worker / Self-Employment (98-10-0592-01)	Priority A — Keep	Self-employment context by available demographic/industry dimensions; proxy only.
CFIB succession research	Supporting reference	Canadian succession-planning context, exit intentions and barriers; not automatically Niagara-specific.
Niagara economic/labour sources	Supporting reference	Local economic, employment and sector context.
2026 NEI succession-specific information	Access pending / high value	Potential direct succession evidence; public access to responses has not been confirmed.

4. Most Important Week 2 Finding

The biggest finding from our Week 2 research is that we have useful data about Niagara businesses, business dynamics, demographics and the wider Canadian succession problem, but we have **not yet confirmed a public Niagara business-level dataset containing the direct succession variables we would ideally want**.

• Owner age

• Retirement intention

• Intended exit timeline

• Identified successor

• Succession-plan status

• Confirmed succession-related reason for closure

> This gap affects how we frame the analysis. We will not hide it, and we will not make a proxy variable look like direct evidence.

5. Rules Agreed for Interpreting the Data

• **Business exit ≠ succession failure.** A business may exit for many reasons.

• **Self-employed ≠ automatically a business owner.** It can only be used as a proxy/context variable where justified.

• **Population age ≠ business-owner age.**

• **Permanently Closed/Vacant ≠ automatically succession-driven closure.**

• A common column name does not automatically mean two datasets can be validly joined.

• Different business-size definitions, NAICS versions, geographies and reference periods must be checked before comparison or integration.

• A variable mentioned in a survey or report will not be recorded as available unless it exists in the actual acquired dataset.

6. Data Source Priorities

Priority A — Core acquisition: datasets selected for Week 3 download and audit.
Priority B — Supporting / Conditional: sources used for context, benchmarking or additional analysis where needed.
Priority C — Access Pending: potentially valuable sources that still require access investigation.

7. Documentation Completed During Week 2

Document	Purpose
docs/DATA_ACQUISITION_PLAN.md	Records selected sources, acquisition priorities, Week 3 actions, decision matrix and source-to-deliverable mapping.
docs/DATA_DICTIONARY.md	Template for documenting actual variables after collection and audit.
data/raw/DATA_SOURCES.md	Raw-data provenance log for source link, filename, date, licence, geography and repository location.
docs/DATA_AUDIT_TEMPLATE.md	Standard Week 3 audit process for all Priority A datasets.
docs/WEEK_2_CONTRIBUTIONS.md	Records team research contributions and consolidated findings.
docs/WEEK_2_HANDOVER.md	Records Week 2 completion and handover into Week 3.
references/DATA_LICENSES.md	Records licensing and usage considerations.
RISK_REGISTER.md	Records major data and project risks identified during research.
README.md	Updated project overview, methodology, structure, deliverables, roadmap and current status.

8. Alignment with Official Challenge Deliverables

No.	Deliverable	How Week 2 supports it
4.1	Aggregated data layer	Selected candidate sources and prepared acquisition, provenance, audit and dictionary documentation.
4.2	Research report	Established evidence sources, limitations, documentation trail and research framing.
4.3	Infographics & visual assets	Identified likely evidence base; final topics depend on analysis.
4.4	Graphs & exploratory visuals	Identified candidate datasets for later EDA.
4.5	Predictive model for forecasting business closures	Documented the unresolved target/label problem. Model feasibility will be determined after data audit.
4.6	Business acquisition / succession playbook	Identified succession research and future analytical evidence base.
4.7	Local-economy impact analysis	Identified employment, business-count, sector and local-economic sources.
4.8	Government & private-sector remediation report	Established the evidence chain that later recommendations must follow.
4.9	Nice-to-have extensions	Extensions are recognised but will not take priority over the core analytical work.

9. Predictive Model — Open Question

The challenge requires a predictive model for forecasting business closures. At the end of Week 2, we have deliberately **not** selected a final algorithm or claimed that we already have a valid succession-risk label. The modelling unit, target variable, forecast horizon, feature set and relationship between general closure and succession-related closure must be resolved from the actual data and project methodology.

> We will not choose a model simply because it is familiar. The model must follow the problem and the evidence.

10. What Week 2 Did Not Include

The following work was intentionally left for later phases and is not considered incomplete Week 2 work:

• Downloading all Priority A datasets

• Cleaning data

• Deleting/filling missing values

• Removing duplicates

• Merging datasets

• Exploratory data analysis

• Feature engineering

• Creating a succession-risk score

• Selecting and training a machine-learning model

• Producing final economic-impact estimates or recommendations

11. Week 3 Handover — Data Collection & Audit

> **Week 3 question:** What data do we actually have, what is inside it, what is its quality, and what can it legitimately support?

The Week 3 workflow will be: **Download** → **Preserve Raw File** → **Record Source** → **Audit** → **Review** → **Cleaning**.

• Download approved Priority A datasets in the preferred available format, with CSV preferred where suitable.

• Preserve the original raw files unchanged.

• Update data/raw/DATA_SOURCES.md immediately after each download.

• Preserve official metadata and definitions.

• Complete the standard data audit for each Priority A dataset.

• Confirm actual rows, columns, data types, geography and time coverage.

• Check missing values and duplicates without cleaning them yet.

• Confirm NAICS/industry and business-size definitions.

• Identify possible joining keys, but do not merge merely because a field name looks similar.

• Check again for direct succession-related variables.

• Continue investigating access to 2026 NEI succession-specific information.

• Update the Risk Register when new problems or gaps are discovered.

12. Week 2 Completion Checklist

All items below were completed as part of the Week 2 documentation process:

• ✓ Major data-source areas researched

• ✓ Niagara local business source identified

• ✓ Statistics Canada sources identified

• ✓ Succession research identified

• ✓ Major data gaps and limitations documented

• ✓ Data licences documented

- ✓ Data acquisition priorities established
- ✓ Week 3 acquisition plan prepared
- ✓ Data dictionary template prepared
- ✓ Raw-data source tracking prepared
- ✓ Data audit template prepared
- ✓ Team contributions documented
- ✓ Sources mapped to challenge deliverables
- ✓ Risks recorded
- ✓ Week 3 handover actions defined
- ✓ README updated to reflect Week 2 completion and Week 3 transition

13. Week 2 Conclusion

Week 2 is complete from a data-source research, evaluation and planning perspective. The week helped us understand both the value and the limitations of the evidence available to us. We identified useful local and national sources, but we also identified a major succession-specific data gap. Instead of forcing the data to answer a question it cannot answer, we have documented the gap and prepared a structured process for testing the actual datasets in Week 3.

> As Team Lead, my main decision at the end of Week 2 is that we are ready to move forward, but we must continue to separate what the data **directly tells us** from what we are only using as a **proxy or context**. This will be important for the credibility of our final analysis, model and recommendations.

Week 2 Status: COMPLETE
Next Phase: Week 3 — Data Collection & Audit
Project Status: In Progress